

# Jacob Helwig

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## Education

Texas A&M University, Ph.D. Computer Science

August 2021-May 2026

The University of Texas at Austin, BS Mathematics

August 2016-May 2021

## Work Experience

LinkedIn, CoreAI Intern

Summer 2024, Summer 2025

- **Summer 2025:** Pretrained transformers for sequential recommendation. Key challenges include correct weight and optimizer state updates with sparse gradients as a result of high percentage of unused embeddings due to large item cardinality/vocab size.
- **Summer 2024:** Developed full pipeline for large-scale training of causal transformers for sequential recommendation. Key challenges include memory management due to large cardinality/vocab size leading to large model head and logits vector. Solved using kernel fusion of model head with cross-entropy loss.

Los Alamos National Laboratory, Applied ML Fellow

Summer 2023

- Project: Solving the Elastic Wave Equation Using Deep Learning

## OSS Contributions

verl, Volcano Engine Reinforcement Learning for LLMs ([All Merged PRs](#))

- **On-Policy Distillation:** vLLM teacher server supporting LLM/VLM distillation with token-in/token-out. *Trainer backends:* FSDP/Megatron. *Distillation losses:* KL estimator/top- $k$  KL with Megatron vocab parallel autograd function. *Optimization algorithm:* policy gradient with negative KL reward/backpropagate distillation loss.
- **Multi-Teacher On-Policy Distillation:** Domain-specific teachers, as used by DeepSeek V4, GLM-5, MiMo-V2-Flash.

## Publications and Talks

|                |                                                                                                                                                                                                                     |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [Preprint]     | <b>Data-Efficient Autoregressive-to-Diffusion Language Models via On-Policy Distillation</b><br>Co-first author                                                                                                     |
| [ICML 2026]    | <b>Learnability-Informed Fine-Tuning of Diffusion Language Models</b><br><a href="#">Paper</a>                                                                                                                      |
| [ICLR 2026]    | <b>A Two-Phase Deep Learning Framework for Adaptive Time-Stepping in High-Speed Flow Modeling</b><br>First author, <a href="#">paper</a> , <a href="#">code</a>                                                     |
| [NeurIPS 2024] | <b>A Geometry-Aware Message Passing Neural Network for Modeling Aerodynamics over Airfoils</b> (talk at the NeurIPS 2024 ML4CFD Competition)<br>Best student solution, <a href="#">paper</a> , <a href="#">code</a> |
| [ICML 2024]    | <b>Equivariance via Minimal Frame Averaging for More Symmetries and Efficiency</b><br>Spotlight, <a href="#">paper</a>                                                                                              |
| [FnT ML]       | <b>Artificial Intelligence for Science in Quantum, Atomistic, and Continuum Systems</b><br>Co-first author, <a href="#">paper</a>                                                                                   |
| [ICLR 2024]    | <b>SineNet: Learning Temporal Dynamics in Time-Dependent PDEs</b><br>Co-first author, <a href="#">paper</a> , <a href="#">code</a>                                                                                  |
| [LoG 2023]     | <b>High-fidelity Fluid Flow Reconstruction</b><br><a href="#">Paper</a>                                                                                                                                             |
| [ICML 2023]    | <b>Group Equivariant Fourier Neural Operators for Partial Differential Equations</b><br>Co-first author, <a href="#">paper</a> , <a href="#">code</a>                                                               |
| [ACM TOMS]     | <b>A Covariate-Dependent Approach to Gaussian Graphical Modeling in R</b><br>First author, <a href="#">paper</a> , <a href="#">R package</a>                                                                        |
| [Preprint]     | <b>An Approximate Bayesian Approach to Covariate-dependent Graphical Modeling</b><br><a href="#">Paper</a>                                                                                                          |

## Honors

Ruth J. & Howard F. Newton Memorial Graduate Student Teaching Award In Statistics, 2022 Recipient

- “Jacob was selected to receive the Newton Teaching Award based on his outstanding evaluations by the instructors he was Teaching Assistant for during the Fall 2021 and Spring 2022 semesters.” ([award details](#))